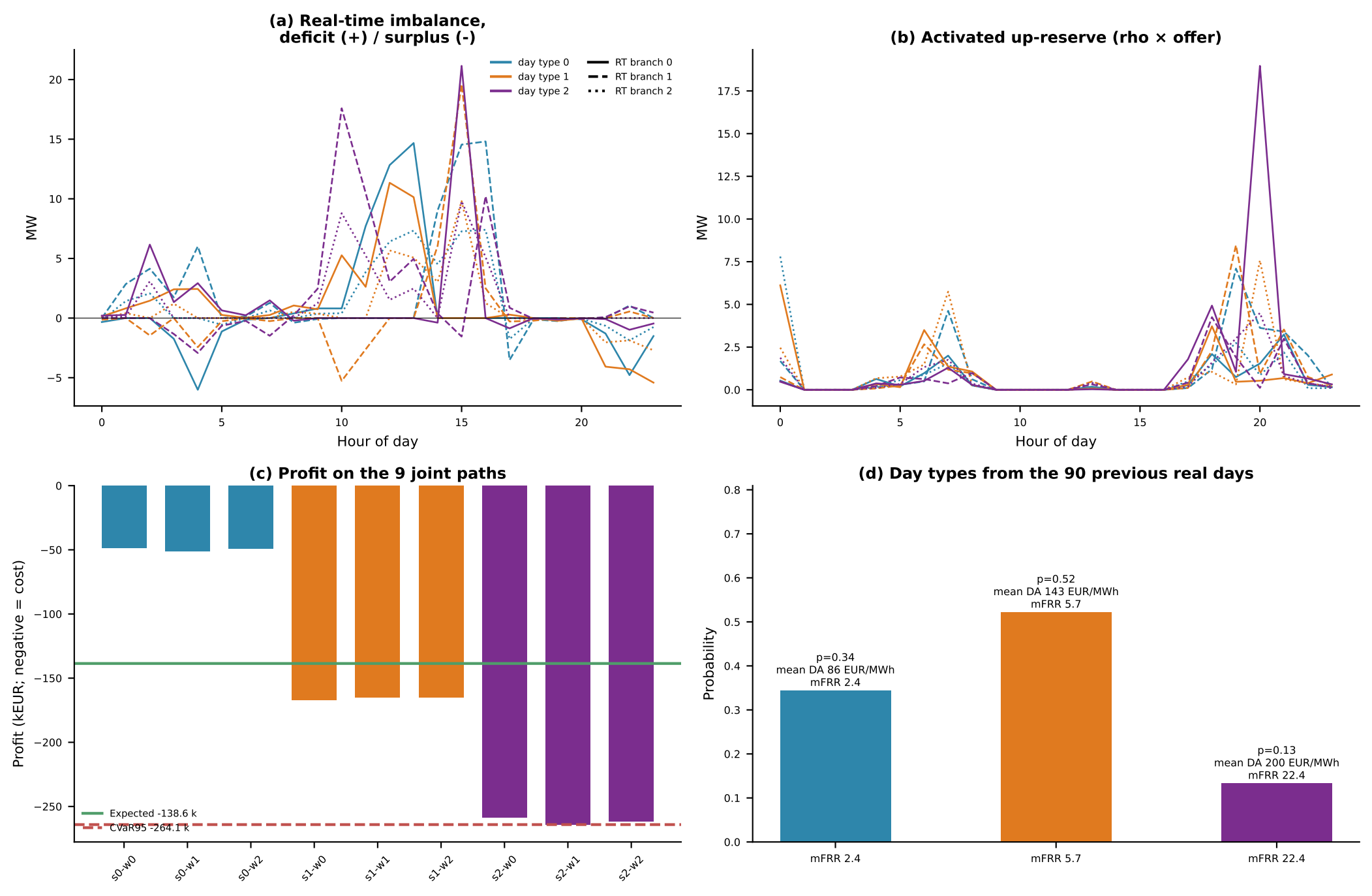
(a) Real-time imbalance, deficit (+) / surplus (-)
day type 0
day type 1
day type 2
RT branch 0
RT branch 1
RT branch 2
MW
Hour of day
(b) Activated up-reserve (rho × offer)
MW
Hour of day
(c) Profit on the 9 joint paths
Profit (kEUR; negative = cost)
Expected -138.6 k
CVaR95 -264.1 k
s0-w0
s0-w1
s0-w2
s1-w0
s1-w1
s1-w2
s2-w0
s2-w1
s2-w2
(d) Day types from the 90 previous real days
Probability
p=0.34
mean DA 86 EUR/MWh
mFRR 2.4
p=0.52
mean DA 143 EUR/MWh
mFRR 5.7
p=0.13
mean DA 200 EUR/MWh
mFRR 22.4
mFRR 2.4
mFRR 5.7
mFRR 22.4